

V.60: Unified Field Theory of Spacetime and Electromagnetism

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V.60: The Geometric Unity of Fields

In the V.60 framework, we propose that the electromagnetic field is not an external influence on the spacetime manifold, but a local manifestation of the manifold's intrinsic curvature. We unify the Einstein field equations with Maxwell's equations through the definition of the "Electromagnetic Metric Tensor" $\mathcal{G}_{\mu\nu}$.

The Unified Metric Representation

We define the unified metric $\mathcal{G}_{\mu\nu}$ as the sum of the gravitational metric $g_{\mu\nu}$ and the electromagnetic potential field A_μ , elevated to a higher-dimensional curvature:

$$\mathcal{G}_{\mu\nu} = g_{\mu\nu} + \kappa(A_\mu A_\nu - \frac{1}{4}g_{\mu\nu}A^\alpha A_\alpha)$$

where κ is the coupling constant between mass-energy and charge density.

Field Equations Integration

By applying the Bianchi identity to this unified metric, we derive a single set of field equations that govern both gravity and electromagnetism simultaneously:

$$\nabla^\mu \mathcal{G}_{\mu\nu} = 4\pi(T_{\mu\nu}^{(matter)} + T_{\mu\nu}^{(em)})$$

In this framework, the Maxwell stress-energy tensor $T_{\mu\nu}^{(em)}$ arises naturally from the asymmetric part of the unified curvature tensor. Thus, the distinction between "geometry" (gravity) and "field" (electromagnetism) vanishes.

Conclusion

The V.60 theory asserts that charges are the "kinks" or "singularities" in the curvature of spacetime. By treating electromagnetic potential as a component of the metric tensor, we achieve a total unification, where the expansion of

the universe and the propagation of light are equivalent processes within the singular manifold. $\square$